For the use only of a Registered Medical Practitioner or Hospital or a Laboratory only xxxxxxxx-5253

ASTHATOR

MONTELUKAST SODIUM TABLETS 4mg/5 mg/10 mg

BRAND OR PRODUCT NAME :
ASTHATOR 4 MG CHEWABLE TABLET
ASTHATOR 5 MG CHEWABLE TABLET
ASTHATOR 10 MG TABLET

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S):

ASTHATOR 4 MG CHEWABLE TABLET
Each chewable tablet contains:
Montelukast sodium 4.16mg eq. to Montelukast 4 mg
Colour: Red Oxide of Iron

ASTHATOR 5 MG CHEWABLE TABLET
Each chewable tablet contains:
Montelukast sodium 5.20mg eq. to Montelukast 5 mg
Colour: Red Oxide of Iron

ASTHATOR 10 MG TABLET
Each uncoated tablet contains:
Montelukast sodium 10.40mg eq. to Montelukast 10 mg
Colour: Red Oxide of Iron & Yellow Oxide of Iron

PRODUCT DESCRIPTION:
ASTHATOR 4: Pink colored, oval biconvex shaped, uncoated tablets, with break line on both the sides.
ASTHATOR 5: Pink colored, round shaped, uncoated tablets, with break line on both the sides.
ASTHATOR 10: Light brown colored, round biconvex shaped, uncoated tablets, with break line on both the sides.

DOSAGE FORM:
Uncoated / Chewable Tablets

CLINICAL PHARMACOLOGY:

Pharmacodynamics:
The cysteinyl leukotrienes (LTC4, LTD4, LTE4) are products of arachidonic acid metabolism and are released from various cells, including mast cells and eosinophils. These eicosanoids bind to cysteinyl leukotriene receptors (CysLT) found in the human airway. Cysteinyl leukotrienes and leukotriene receptor occupation have been correlated with the pathophysiology of asthma, including airway edema, smooth muscle contraction, and altered cellular activity associated with the inflammatory process, which contribute to the signs and symptoms of asthma.
Montelukast is an orally active compound that binds with high affinity and selectivity to the CysLT1 receptor (in preference to other pharmacologically important airway receptors, such as the prostanoid, cholinergic, or (beta) -adrenergic receptor). Montelukast inhibits physiologic actions of LTD4 at the CysLT1 receptor without any agonist activity.

Pharmacokinetics:

Absorption:
Montelukast is rapidly absorbed following oral administration.After administration of the 10 mg tablet to fasted adults, the mean peak plasma concentration (C_{max}) is achieved three to four hours (T_{max}). The mean oral bioavailability is 64%. The oral bioavailability and C_{max} are not influenced by a standard meal in the morning.
For the 5 mg tablet, the mean Cmax is achieved in 2 to 2.5 hours after administration to adults in the fasted state. The mean oral bioavailability is 73% in the fasted state verses 63 % when administered with a standard meal in the morning.
After administration of the 4 mg tablet to pediatric patients 2 to 5 years of age in the fasted state, C_{max} is achieved 2 hours after administration. The mean C_{max} is 66% higher while mean C_{min} is lower than in adults receiving a 10 mg tablet.
The safety and efficacy of ASTHATOR were demonstrated in clinical trials in which both formulations were administered in the evening without regard to the timing of food ingestion.

Distribution:
Montelukast is more than 99% bound to plasma proteins. The steady-state volume of distribution of montelukast averages 8-11 liters. Studies in rats with radiolabeled montelukast indicate minimal distribution across the blood - brain barrier. In addition, concentrations of radiolabeled material at 24 hours post-dose were minimal in all other tissues.

Metabolism:
Montelukast is extensively metabolised. In studies with therapeutic doses, plasma concentrations of metabolites of montelukast are undetectable at steady state in adults and pediatric patients.
In vitro studies using human liver microsomes indicate that cytochromes P450 3A4, 2A6 and 2C9 are involved in the metabolism of montelukast. Based on further in vitro results in human liver microsomes, therapeutic plasma concentrations of montelukast do not inhibit cytochromes P450 3A4, 2C9, 1A2, 2A6, 2C19, or 2D6.

Elimination:
The plasma clearance of montelukast averages 45 ml/min in healthy adults. Following an oral dose of radiolabeled montelukast, 86% of the radioactivity was recovered in 5-day faecal collections and < 0.2% was recovered in urine. Coupled with estimates of montelukast oral bioavailability, this indicates that montelukast and its metabolites are excreted almost exclusively via the bile. In several studies, the mean plasma half-life of montelukast ranged from 2.7 to 5.5 hours in healthy young adults. The pharmacokinetics of montelukast is nearly linear for oral doses up to 50 mg. During once-daily dosing with 10-mg montelukast, there is little accumulation of the parent drug in plasma (~14%).

Special Populations:

Gender
The pharmacokinetics of Montelukast is similar in males and females.

Elderly
The pharmacokinetic profile and the oral bioavailability of a single 10-mg oral dose of Montelukast are similar in elderly and younger adults. The plasma half-life of Montelukast is slightly longer in the elderly. No dosage adjustment in the elderly is required.

Race
Pharmacokinetic differences due to race have not been studied.

Hepatic Insufficiency
Patients with mild-to-moderate hepatic insufficiency and clinical evidence of cirrhosis had evidence of decreased metabolism of Montelukast resulting in 41% (90% CI=7%, 85%) higher mean Montelukast area under the plasma concentration curve (AUC) following a single 10-mg dose. The elimination of Montelukast was slightly prolonged compared with that in healthy subjects (mean half-life, 7.4 hours). No dosage adjustment is required in patients with mild-to-moderate hepatic insufficiency. The pharmacokinetics of Montelukast in patients with more severe hepatic impairment or with hepatitis has not been evaluated.

Renal Insufficiency
Since Montelukast and its metabolites are not excreted in the urine, the pharmacokinetics of Montelukast was not evaluated in patients with renal insufficiency. No dosage adjustment is recommended in these patients.

Adolescents and Pediatric Patients
The plasma concentration profile of Montelukast following administration of the 10-mg tablet is similar in adolescents (>/=) 15 years of age and young adults. The 10-mg tablet is recommended for use in patients (>/=) 15 years of age.

INDICATIONS:
For the prophylaxis and chronic treatment of asthma in adults and pediatric patients 2 years of age and older.

ASTHATOR is indicated in adults and pediatric patients 2 years of age and older for the relief day time and night time symptoms of allergic rhinitis.

RECOMMENDED DOSAGE:
ASTHATOR should be taken once daily. For asthma, the dose should be taken in the evening. For allergic rhinitis, the time of administration may be individualized to suit patient needs. Patients with both asthma and allergic rhinitis should take only one tablet daily in the evening.
<u>Adults and Adolescents 15 Years of Age and Older with Asthma or Allergic Rhinitis</u>
The dosage for adults and adolescents 15 years of age and older is one 10-mg tablet daily.
<u>Pediatric Patients 6 to 14 Years of Age with Asthma or Allergic Rhinitis</u>
The dosage for pediatric patients 6 to 14 years of age is one 5-mg tablet daily. No dosage adjustment within this age group is necessary
<u>Pediatric Patients 2 to 5 Years of Age with Asthma or Allergic Rhinitis</u>
The dosage for pediatric patients 2 to 5 years of age is one 4-mg tablet daily.

MODE OF ADMINISTRATION: Oral

CONTRAINDICATIONS:
Hypersensitivity to the active substance or to any of the excipients.

WARNINGS AND PRECAUTIONS:

"Unsuitable for phenylketonurics."

Patients should be advised never to use oral montelukast to treat acute asthma attacks and to keep their usual appropriate rescue medication for this purpose readily available. If an acute attack occurs, a short-acting inhaled β-agonist should be used. Patients should seek their doctor's advice as soon as possible if they need more inhalations of short-acting β-agonists than usual.
Montelukast should not be abruptly substituted for inhaled or oral corticosteroids. There are no data demonstrating that oral corticosteroids can be reduced when montelukast is given concomitantly.
In rare cases, patients on therapy with anti-asthma agents including montelukast may present with systemic eosinophilia, sometimes presenting with clinical features of vasculitis consistent with Churg-Strauss syndrome, a condition which is often treated with systemic corticosteroid therapy. These cases usually, but not always, have been associated with the reduction or withdrawal of oral corticosteroid therapy. The possibility that leukotriene receptor antagonists may be associated with emergence of Churg-Strauss syndrome can neither be excluded nor established. Physicians should be alert to eosinophilia, vasculitic rash, worsening pulmonary symptoms, cardiac complications, and/or neuropathy presenting in their patients. Patients who develop these symptoms should be reassessed and their treatment regimens evaluated.
ASTHATOR contains aspartame, a source of phenylalanine: hence it should be prescribed cautiously to the patients with phenyketonuria.

INTERACTIONS WITH OTHER MEDICAMENTS:
Montelukast may be administered with other therapies routinely used in the prophylaxis and chronic treatment of asthma. In drug-interactions studies, the recommended clinical dose of montelukast did not have clinically important effects on the pharmacokinetics of the following drugs: theophylline, prednisone, prednisolone, oral contraceptives (ethinyl oestradiol/norethindrone 35/1), terfenadine, digoxin and warfarin.
The area under the plasma concentration curve (AUC) for montelukast was decreased approximately 40% in subjects with co-administration of phenobarbital. Since montelukast is metabolised by CYP 3A4, caution should be exercised, particularly in children, when montelukast is co-administered with inducers of CYP 3A4, such as phenytoin, phenobarbital and rifampicin.
In vitro studies have shown that montelukast is a potent inhibitor of cyp 2c8. However, data from a clinical drug-drug interaction study involving montelukast and rosiglitazone (a probe substrate representative of drugs primarily metabolised by cyp 2c8) demonstrated that montelukast does not inhibit cyp 2c8 in vivo. Therefore, montelukast is not anticipated to markedly alter the metabolism of drugs metabolised by this enzyme (eg., paclitaxel, rosiglitazone, and repaglinide).

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION:

PREGNANCY CATEGORY B:
ASTHATOR has not been studied in pregnant women. ASTHATOR should be used during pregnancy only if clearly needed.

NURSING MOTHERS
It is not known If ASTHATOR is excreted in human milk. Because many drugs are excreted in human milk, caution should be exercised when ASTHATOR is given to a nursing mother.

PEDIATRIC USE
ASTHATOR has been studied in pediatric patients 6 months to 14 years of age. Safety and effectiveness in pediatric patients younger than 6 months of age have not been studied.

ADVERSE EFFECTS / UNDESIRABLE EFFECTS:
The following adverse reactions have been reported in post-marketing use:
Blood and lymphatic system disorders: increased bleeding tendency, thrombocytopenia.
Immune system disorders: hypersensitivity reactions including anaphylaxis, hepatic eosinophilic infiltration.
Psychiatric disorders: dream abnormalities including nightmares, hallucinations, insomnia, irritability, anxiety, restlessness, agitation including aggressive behaviour, tremor, depression, suicidal thinking, behaviour (suicidality) in very rare cases and obsessive-compulsive symptoms.
Nervous system disorders: dizziness drowsiness, paraesthesia/hypoesthesia, seizure.
Cardiac disorders: palpitations.
Gastro-intestinal disorders: diarrhoea, dry mouth, dyspepsia, nausea, vomiting.
Hepatobiliary disorders: elevated levels of serum transaminases (ALT, AST), cholestatic hepatitis
Skin and subcutaneous tissue disorders: angiooedema, bruising, urticaria, pruritus, rash, erythema nodo sum.
Musculoskeletal and connective tissue disorders: arthralgia, myalgia including muscle cramps
General disorders and administration site conditions: asthenia/fatigue, malaise, oedema.
Very rare cases of Churg-Strauss Syndrome (CSS) have been reported during montelukast treatment in asthmatic patients.

OVERDOSE AND TREATMENT:
No specific information is available on the treatment of overdosage with montelukast. In chronic asthma studies, montelukast has been administered at doses up to 200 mg/day to adult patients for 22 weeks and in short-term studies, up to 900 mg/day to patients for approximately one week without clinically important adverse experiences.
There have been reports of acute overdosage in post-marketing experience and clinical studies with montelukast. These include reports in adults and children with a dose as high as 1000 mg (approximately 61 mg/Kg in a 42 month old child). The clinical and laboratory findings observed were consistent with the safety profile in adults and pediatric patients. There were no adverse experiences in the majority of overdosage reports. The most frequently occurring adverse experiences were consistent with the safety profile of montelukast and included abdominal pain, somnolence, thirst, headache, vomiting, and psychomotor hyperactivity.
It is not known whether montelukast is dialyzable by peritoneal- or haemo-dialysis.

STORAGE:
STORE BELOW 30^{o}C, PROTECTED FROM MOISTURE.

EXPIRY DATE:
Do not use later than the date of expiry.

DATE OF REVISION OF PACKAGE INSERT:
13th November, 2019

PRESENTATION
ASTHATOR TABLETS are packed in Alu - Alu blisters of 10 tablets. Such blisters containing 10 tablets are packed into a carton of 30's and 100's.
"Not all presentations may be available locally"

NAME AND ADDRESS OF MANUFACTURER:

torrent PHARMA

TORRENT PHARMACEUTICALS LTD.
Indrad-382 721, Dist. Mehsana, INDIA.

PRODUCT NAME	:	Asthator	**COUNTRY :** Malaysia	**LOCATION :** Indrad		**Supersedes A/W No.:**		
ITEM / PACK	:	Insert	**NO. OF COLORS:** 1	**REMARK :**				
DESIGN STYLE	:	Front/Back	**PANTONE SHADE NOS.:**	**SUBSTRATE :**				
CODE	:	xxxxxxxxxx-5253	Black	**Activities**	**Department**	**Name**	**Signature**	**Date**
DIMENSIONS (MM)	:	150 x 240 mm		Prepared By	**Pkg.Dev**			
ART WORK SIZE	:	S/S		Reviewed By	**Pkg.Dev**			
DATE	:	13-11-2019		Approved By	**Quality**			

This colour proof is not colour binding. Follow Pantone shade reference for actual colour matching.